Enrolment No.:

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INSTITUTE OF INFORMATION TECHNOLOGY
ADDING VALUE TO ENGINEERING

Total No. of Questions: 8] **[Total No. of Printed Pages: 3**

BTECH_ET_SEM5 / ET Fifth Semester
ETPCC502 / Electromagnetic Theory
(Nov. 2025, 2025/2024 Pattern)

Time: 2:30 Hrs. **Maximum Marks: 60**

Instructions: -

(i) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
(ii) Neat diagrams must be drawn wherever necessary.
(iii) Figures to the right indicate full marks.
(iv) Assume suitable data, if necessary.
(v) Only Scientific calculators are allowed.

Q1. A) Using Biot Savart law, find H due to infinitely long straight filament current of I amperes. **[7] CO3 [UNDERSTANDING]**

B) A point charge Q=-60nC, is moving with a velocity $6*10^6$ m/s in the direction specified by unit vector -0.48ax-0.6ay+0.64az. Find the magnitude of the force on a moving charge in the magnetic field, B=2ax-6ay+5az mT. **[8] CO3 [UNDERSTANDING]**

(OR)

Q2. A) Write a short note on Lorentz force equation. A charge of 10 nC is moving with velocity of 10^7 (-0.5 $a_x + a_y$ - 0.71 a_z) m/s. Determine the force exerted on the test charge when an electric field E = (3 a_x +2 $a_y + a_z$) kV/m is applied. **[7] CO3 [UNDERSTANDING]**

B) Find the flux passing the region of plane $\phi = \pi/4$ defined by $0.01 \leq r \leq 0.05$m and $0 \leq z \leq 2$m.A current filament of 2.5 A is along the z axis along a_z direction in free space. **[8] CO3 [UNDERSTANDING]**

Q3. A) Derive the boundary conditions for electric field at an interface between conductor and free space **[7] CO4 [UNDERSTANDING]**

B) Let εr1 = 2.5 for 0< y <1mm, εr2= 4 for 1< y<3mm and εr3 for 3< y < 5mm.Conducting surfaces are present at y = 0 and y = 5mm. Calculate the capacitance per square meter of surface are if a) εr3 is that of air b) εr3 = εr1 c) εr3 = εr2 d) if region 3 is silver. **[8] CO4 [UNDERSTANDING]**

(OR)

Q4. A) Derive the boundary conditions for electric field at an interface between two different dielectrics. **[7] CO4 [UNDERSTANDING]**

B) Derive the expression for capacitance of co-axial cable.Also find the capacitance of 20cm coaxial cable aving an inner conductor with 0.0295 inches diameter an an outer conductor with inside diameter of 0.116 inches with εr = 2.26. **[8] CO4 [UNDERSTANDING]**

Q5. A) Define displacement current and displacement current density and hance show that $\Delta \times H = Jc + Jd$ **[7] CO5 [UNDERSTANDING]**

B) The conduction current flowing through a copper wire of cross-sectional area $2*10^{-5}$ $*m^2$ is 3 A if μ = $4\pi*10^7$* H/m ε = 8.854 * 10^{-12} and σ = 5.8 * 10^7 ʊ / m Find the displacement current at f = 1GHz **[8] CO5 [UNDERSTANDING]**

(OR)

Q6. A) State poynting theorem and significance of poynting vector. **[7] CO5 [UNDERSTANDING]**

B) In charge free region the magnetic field intensity is given by, $H = H_m \cos ßz \cos \omega t \, \bar{a}_y$ By using Maxwell's equation calculate electric field intensity. **[8] CO5 [UNDERSTANDING]**

Q7. A) What is polarization of uniform plane waves? Write a short note on a) Linear polarization b) Elliptical polarization c) Circular polarization **[7] CO6 [APPLYING]**

B) Calculate the skin depth and wave velocity at 2MHz in aluminium with conductivity 40 MS/m and μr = 1. **[8] CO6 [APPLYING]**

(OR)

Q8. A) Derive Helmholtz equations using maxwell's equations for E and H fields. **[7] CO6 [APPLYING]**

B) A lossy dielectric is characterized by $\varepsilon r = 2.5$, $\mu r = 4$ and $\sigma = 10^{-3}$ mho/m at frequency 10 MHz. Find a)Attenuation constant b)Phase constant c) Velocity of propagation d) Wavelength e) Intrinsic impedance **[8] CO6 [APPLYING]**

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